Christian Giancola

MECHANICAL ENGINEERING · ROBOTIC CAM · ENGINEERING SOFTWARE

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PROFILE

Mechanical Engineering BSc candidate focused on robotic additive manufacturing, computational geometry and engineering software. I develop multi-axis toolpaths in Rhino and Grasshopper and Python postprocessors for KRL generation, coordinate transformations, tool orientation, extrusion control and external-axis coordination. I want to apply this combination of manufacturing knowledge and software development to automated CAD, CAM and robotic process-planning workflows. I founded Orbinel Robotics, where I am designing Kheiron, a 6-axis desktop robotic arm, and its open-source simulator.

EXPERIENCE

Founder & CEO | Orbinel Robotics · orbinel.it **2026 – Present**

Early-stage robotics startup, Bologna

- Founded and lead a modular, open robotics startup: 3D-printable robots, components with a standard mounting interface, an open-source simulator and a robotics course for schools.
- Designing its first robot, Kheiron (a 6-axis desktop arm), and developing Orbinel Motion, the software that simulates and drives it.
- Coordinating a team of three engineering students covering robot applications, C++ firmware and controls, assembly, testing and FEM checks.
- Bilingual website with a waiting list and newsletter, GDPR-compliant forms and a parametric tool generator that runs in the browser.

Industrial Robotics and Additive Manufacturing Intern | University of Bologna **Mar 2026 – Oct 2026**

Montecuccolino Laboratory, Bologna

- Developed software-driven toolpaths for robotic material deposition using a six-axis KUKA robot, Rhino 8, Grasshopper and KUKA|prc.
- Designed planar, inclined and non-planar slicing strategies for complex surfaces and multi-axis trajectories.
- Developed Python postprocessors that convert geometric path data into KRL programs and manage coordinate transformations, tool orientation, extrusion, an external linear axis, homing and purge sequences.
- Integrated, configured and tested a system comprising the robot, extrusion unit, tool and external linear axis.
- Built parametric geometry-processing workflows for curve ordering, path continuity, surface-normal control and local tool-frame generation.
- Simulated and checked reachability, singularities, orientation and trajectories before execution on the robot.
- Performed hands-on printing tests and iteratively modified geometry, process parameters and code based on observed results.

PROJECTS

Kheiron, 6-axis desktop robotic arm | Orbinel Robotics · orbinel.it/kheiron **2026 – Present**

- Designed from scratch with a spherical wrist: complete CAD generated by Python scripts through the Fusion API, 3D-printable parts, two shoulder servos for the required torque.
- Feetech STS3215 serial-bus servos with an ESP32 bridge whose firmware handles the emergency stop on its own, even if the computer hangs.
- Closed-form forward and inverse kinematics, checked on 2000 random poses; first servo tested and measured on the bench.

Orbinel Motion | Open-source simulator · orbinel.it/motion **2026**

- Browser simulator (JavaScript, Three.js, Web Serial) with kinematics, joint-limit and collision checks, gravity torque on every joint and workspace analysis; drives the real robot over USB.
- Six lessons for schools with self-checking exercises, and interchangeable tools mounted on the robot flange.

KinePath | Personal software project · chrigia12.github.io/KinePath **2026**

- TypeScript web application that imports meshes and BREPs, generates planar and non-planar toolpaths, checks kinematics and collisions for the KUKA KR16 and exports KRL programs. Developed from laboratory process-planning requirements.

Toolpath Check | Personal software project **2026**

- Browser-based tool for analysing and validating robotic paths through geometric checks, program inspection and workflow automation.

FrameForge | Personal software project **2026**

- Browser-based platform for generating usable CAD geometry from images, videos and mechanical drawings.

EDUCATION

BSc in Mechanical Engineering | University of Bologna **2023 – Present**

- Experimental thesis, in progress, on the generation and control of toolpaths for robotic additive manufacturing.
- Main areas: mechanics, mechanical design, manufacturing technology, automation, programming and industrial systems.

First-level Academic Diploma Programme in Classical Guitar | G. B. Martini Conservatory, Bologna **2024 – Present**

TECHNICAL SKILLS

Automation and robotics: KUKA KRL, KUKA|prc, offline programming, simulation, six-axis robots, external axes, forward and inverse kinematics

Robotic CAM: robotic deposition, planar and non-planar slicing, multi-axis toolpaths, tool orientation, collision-aware planning, technical validation

Software development: Python, TypeScript, JavaScript, C++, Three.js, Web Serial, postprocessing, geometric-data processing, automated testing, browser-based engineering tools

Computational design: Rhino 8, Grasshopper, curves, surfaces, BREPs, meshes, data trees, computational geometry

CAD and drafting: PTC Creo, AutoCAD, Autodesk Fusion (Python API), 3D modelling, technical drawings

Electronics: ESP32 (Arduino), serial-bus servos, UART, emergency stop

Engineering methods: prototyping, machine testing, troubleshooting, technical documentation, iterative improvement

LANGUAGES AND ADDITIONAL INFORMATION

Italian: native · **English:** B2, Cambridge First Certificate (Grade C), used for technical documentation and communication

Italian and EU citizen · Driving licence B · Basketball referee · Available from February 2027 for short technical placements and engineering software projects